

The empirical recurrence for OEIS A280158, recovered from its tail

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A280158 records “Empirical recurrence of order 70” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 910$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 11$, so the recurrence is proved for every $n \geq 81$. Its last coefficient is $c_{70} = -41990400 \neq 0$, so no further tail satisfies anything shorter; any order-70 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A280158 is “Number of $n \times 5$ 0..1 arrays with no element unequal to a strict majority of its horizontal, vertical and antidiagonal neighbors, with the exception of exactly one element, and with new values introduced in order 0 sequentially upwards.”. It has offset 1 and begins

5, 40, 305, 1854, 10677, 58852, 318220, 1695030, . . .

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Dec 27 2016 09:20:57, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 910$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 910$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 70: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 11$ is the first whose minimal order is exactly the stated 70.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1820$ exact terms from $n = 11$ on, it returns order 70 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	16	c_{19}	−258268306	c_{37}	69245949760	c_{55}	143470416448
c_2	−72	c_{20}	−118078627	c_{38}	261911792464	c_{56}	−78408688288
c_3	−130	c_{21}	853066098	c_{39}	−384055091016	c_{57}	−13711407792
c_4	1888	c_{22}	−356766843	c_{40}	100554799264	c_{58}	55708266704
c_5	−2914	c_{23}	−1992699176	c_{41}	330024204644	c_{59}	−44755508816
c_6	−15571	c_{24}	2573420777	c_{42}	−492464647672	c_{60}	9926390192
c_7	53930	c_{25}	2699072064	c_{43}	197062743624	c_{61}	13637459040
c_8	33712	c_{26}	−8279462492	c_{44}	283370665732	c_{62}	−15377768224
c_9	−388582	c_{27}	651008434	c_{45}	−499939379672	c_{63}	6829837632
c_{10}	318903	c_{28}	16965146551	c_{46}	300208673692	c_{64}	1006719872
c_{11}	1522002	c_{29}	−13899630322	c_{47}	136037513168	c_{65}	−3188426560
c_{12}	−3217561	c_{30}	−22681254977	c_{48}	−410461171804	c_{66}	1773460864
c_{13}	−2812310	c_{31}	43193530356	c_{49}	313464592088	c_{67}	−279448320
c_{14}	15447649	c_{32}	12558211068	c_{50}	−9423374748	c_{68}	−285697936
c_{15}	−4496540	c_{33}	−96150415336	c_{51}	−241401644648	c_{69}	206789760
c_{16}	−48849845	c_{34}	60302275236	c_{52}	251544279564	c_{70}	−41990400
c_{17}	55832756	c_{35}	117518918988	c_{53}	−75140210448		
c_{18}	107683412	c_{36}	−228865129708	c_{54}	−91898388224		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 81$, and it is the recurrence OEIS A280158 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 11$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{70} - c_1 x^{70-1} - \dots - c_{70}$. Its constant term is $-c_{70} = 41990400$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination

of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 70 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 70, so $\deg q = 70$, and it is monic. It holds from some index t on. From $\max(t, 11)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 70, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 70, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -41990400 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-70 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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